SIMATS ENGINEERING
ASSESSMENT TOOL

COURSE NAME: Theory of Computation COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: De**ign Finite Automata DFA, NFA, ε-NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)**

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:25 Duration : 1 Hour

Design Task

Code of Conduct:

I ____U.Divya/192372277____ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result

in disciplinary action. ***Signature of the student:*** *Divya*

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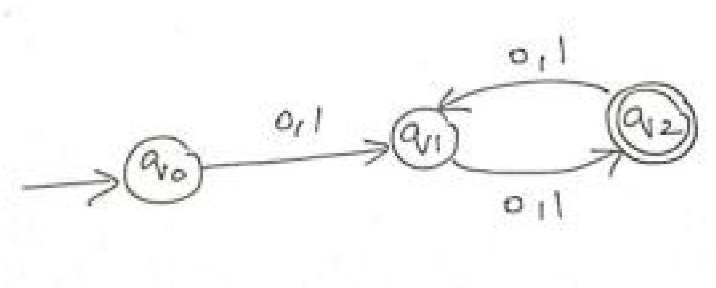
1. Design the regular expression for the language represented by the given automata



Given DFA

- Start state: **q0**
- q0 → q1 on **0 or 1**
- q1 ↔ q2 on **0 or 1**
- q2 is the only final state.

Observation

- The first symbol takes the automaton from q0 to q1.
- Every additional symbol alternates between q1 and q2.
- Acceptance occurs when the total number of symbols is **even**.

Hence, the language consists of **all binary strings of even length**.

Regular Expression

$$((0+1)(0+1))^*$$

or equivalently,

$$((0 \mid 1)(0 \mid 1))^*$$

(If the empty string is not considered, then use)

$$(0 \mid 1)(0 \mid 1)((0 \mid 1)(0 \mid 1))^*$$

2.Consider the following grammar

S->aSc|T|U|e

T-> bTc|e

U->aUb|e

A} What is the language defined by the grammar?

B)Give the derivation tree for the word aaabbc

C)A Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

Consider the Grammar

S->aSc|T|U|e

T-> bTc|e

U->aUb|e

A) Language Defined by the Grammar

The grammar generates three types of strings.

From

$$S \rightarrow aSc$$

equal numbers of **a's** and **c's**

Example

- ε
- ac
- aacc
- aaaccc

From

$$T \rightarrow bTc$$

equal numbers of **b's** and **c's**

Example

- ε
- bc
- bbcc
- bbbccc

From

$$U \rightarrow aUb$$

equal numbers of **a's** and **b's**

Example

- ε
- ab
- aabb
- aaabbb

Therefore,

$$L=\{ancn\} \cup \{bncn\} \cup \{anbn\}, n \geq 0$$

B) Derivation Tree for aaabbc

Derivation

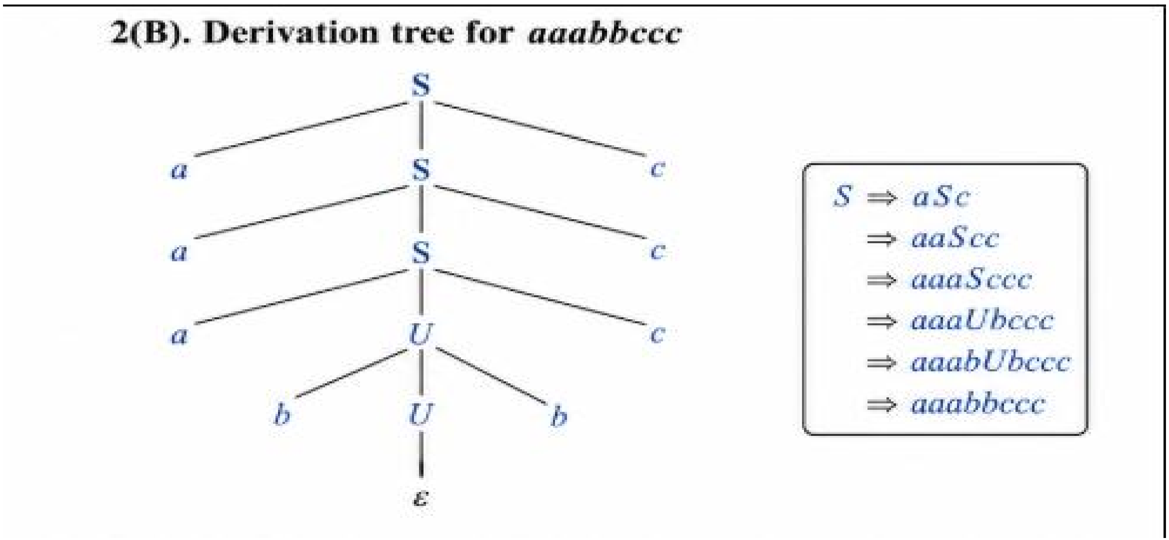
S
⇒ aSc
⇒ aaScc
⇒ aaaSccc
⇒ aaaUbccc
⇒ aaabUbccc
⇒ aaabbccc

The string generated is

aaabbccc

Hence, **aaabbc cannot be derived** from the given grammar because the grammar always produces balanced numbers of symbols.

Answer: aaabbc is not in the language. Therefore, no derivation tree exists.



C) Is the Grammar Ambiguous? Justify

Yes.

The grammar is **ambiguous** because the empty string (ε) can be derived in multiple ways.

Example

$S \rightarrow \varepsilon$

or

$S \rightarrow T \rightarrow \varepsilon$

or

$S \rightarrow U \rightarrow \varepsilon$

Thus, the same string has more than one derivation.

Therefore, the grammar is ambiguous.

3. Design a non deterministic finite automata with ε-transitions to represent the language denoted by the regular expression
00*1+11*0

Regular Expression

00* 1+11* 000^*1+11^*000* 1+11* 0

This means

0+1

after one or more leading 0's

OR

1+0

after one or more leading 1's.

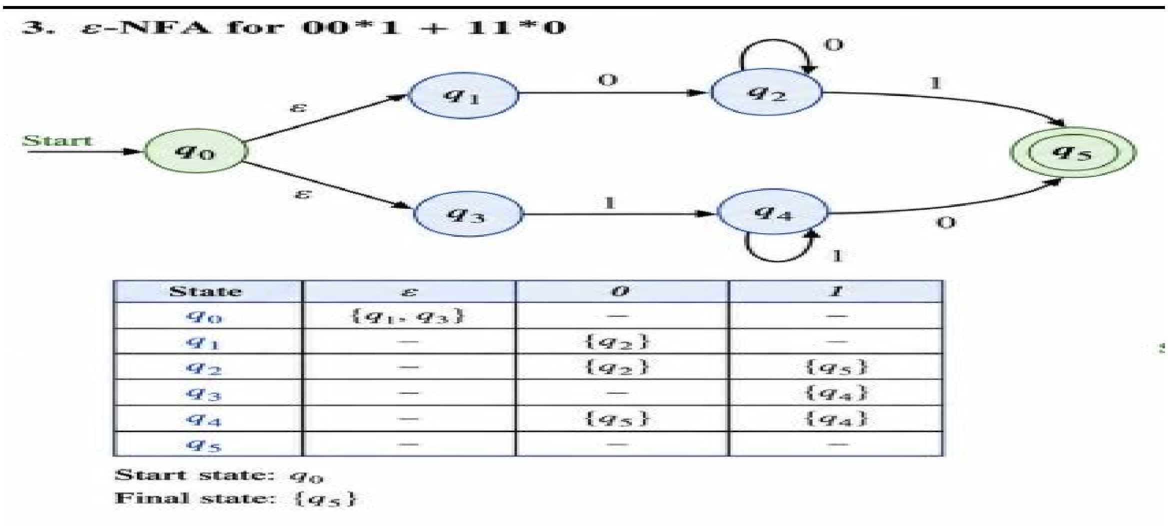
Equivalent forms

0(0)*1

OR

1(1)*0

ε-NFA



State	ε	0	1
q_0	$\{q_1, q_3\}$	—	—
q_1	—	$\{q_2\}$	—
q_2	—	$\{q_2\}$	$\{q_5\}$
q_3	—	—	$\{q_4\}$
q_4	—	$\{q_5\}$	$\{q_4\}$
q_5	—	—	—

Start state: q_0
Final state: $\{q_5\}$

Transition table

State	0	1	ε
q0	-	-	q1,q3

q1	q2	-	-
q2	q2	q5	-
q3	-	q4	-
q4	q5	q4	-
q5	-	-	-

4. **I)**Show that if L1 and L2 are regular then (L1+l2) also regular

Since L_1 and L_2 are regular languages, there exist DFAs

- M_1 accepting L_1
- M_2 accepting L_2.

Construct an NFA by adding a new start state with ε-transitions to the start states of M_1 and M_2.

The new automaton accepts every string accepted by either M_1 or M_2.

Hence,

$$L1+L2=L1 \cup L2$$

is accepted by a finite automaton.

Therefore,

$L_1 + L_2$ is regular.

Ii)Is the language { a^p| p is prime} regular? Justify

No.

Using the Pumping Lemma,

Assume the language is regular.

Choose a sufficiently long prime length string.

After pumping, the resulting string length becomes composite.

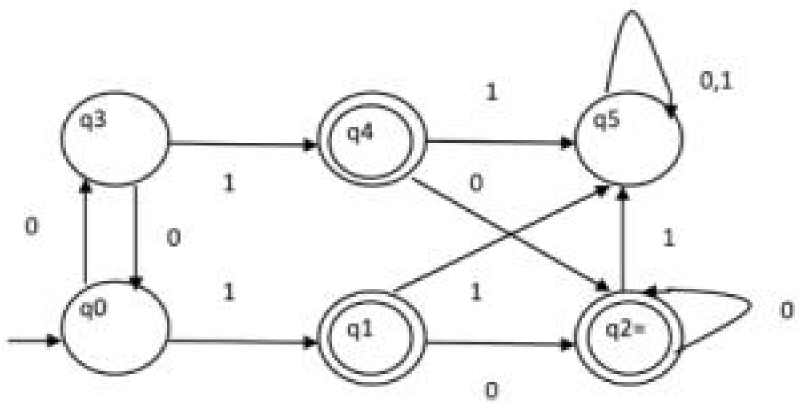
Hence the pumped string no longer belongs to the language.

This contradicts the Pumping Lemma.

Therefore,

{ap | p is prime} is NOT regular.

5. Design minimized automata, Consider the DFA given below. Find the states that are equivalent and construct a minimum state automata equivalent to the given automata. Draw the minimized automata, write its transition table and mark initial and final states



Final States

- q1
- q2
- q4

Non-final States

- q0
- q3
- q5

Equivalent States

Checking transitions,

- q1 and q4 behave identically.
- q2 is different because it has a self-loop on 0.
- q0, q3 and q5 are distinguishable.

Therefore,

$$q1 \equiv q4$$

Minimized States

Let

A = {q0}
B = {q3}
C = {q1,q4}
D = {q2}
E = {q5}

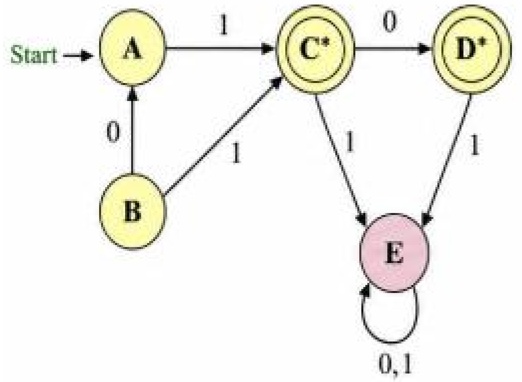
Transition Table

State	**0**	**1**	**Final**
A	B	B	No
B	A	C	No
C	D	E	Yes
D	D	E	Yes
E	E	E	No

Initial State: **A**

Final States: **C, D**

5. Minimized DFA



State	0	1	Final
$A = \{q_0\}$	B	C	No
$B = \{q_3\}$	A	C	No
$C = \{q_1, q_4\}$	D	E	Yes
$D = \{q_2\}$	D	E	Yes
$E = \{q_5\}$	E	E	No

Start state: A

Final states: C, D

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand